Ranjit Sapkota

Data Science Graduate with specialization in Machine Learning

sapkota.ranjit@gmail.com | 0450 654 018 | linkedin.com/in/sapkota-ranjit/

Professional Summary

Data Science graduate and PhD candidate specializing in Machine Learning and Applied Mathematics, with hands-on experience in predictive modelling, time-series forecasting, and data visualization using Python, TensorFlow, SQL, and Power BI. Strong analytical and communication skills with proven ability to turn complex data into actionable insights. Passionate about applying data-driven analytics, digital innovation, and optimization modelling to support energy system forecasting, market analysis, and operational reliability within AEMO's dynamic energy environment.

Technical Skills

• **Languages & Frameworks**: Python, R, SQL, C++, TensorFlow, Pandas
• **Data Visualization & Reporting**: Power BI, Tableau, Excel (Power Query, Power Pivot)
• **Software & Tools**: Machine Learning, Statistical Analysis, Automation, Cybersecurity Analysis

Education

Doctor of Philosophy (PhD), Information Technology - Machine Learning

- **Federation University Australia | 2025 - 2028 (expected)**
- **Research focus: advanced machine learning for large-scale modelling for disease classification**
- **Industry partner: Wills Eye Hospital (USA)**
- **Awards**: Wills Eye Health Stipend ($38,000 AUD per annum), RTP Fee Offset ($27,300 AUD per annum)

Master of Data Science, Federation University, Ballarat, VIC, Australia (Nov 2024)

- **Relevant Coursework**: Machine Learning and Artificial Intelligence, Cyber Threat Intelligence, Data Structures and Algorithms, Data Analytics, Experiment Design & Visualization, Regression and Multivariate Data Analysis, Computational Methods for Data Analysis (Optimization Techniques and Algorithms)
- **Awards**: Federation University Australia Global Excellence Scholarship recipient, Academic Excellence Commendation from the Dean's office, CGPA: 6.563/7

BSc Applied Mathematics, University of Cincinnati, Cincinnati, Ohio, USA (Apr 2020)

- **Relevant Coursework**: Mathematical Modeling, Regression Analysis, Experimental Design & Analysis, Partial Differential Equations
- **Awards**: Mary Jane Toepfer Memorial Award, UC Global Scholarship recipient, Dean's List, CGPA: 3.061/4

Technical Projects & Research Experience

Comorbidity & Disease Detection via Latent Representations, Lead Researcher & ML Engineer (Nov 2024)

- Developed an advanced ML pipeline for automated ophthalmic disease classification using OCT data.
- Engineered U-Net segmentation for RNFL thickness maps, implemented Beta-VAE for feature extraction, and trained a multi-label MLP classifier.
- Achieved classification accuracy of 69%, macro-AUC of 0.838; validated distinct clustering with t-SNE.
- Authored comprehensive research paper detailing methods and implications.
- Tools: Python, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib

Predictive Model for Glioblastoma Survival, Machine Learning Researcher (Feb 2025)

- Developed machine learning model (PLSDA) predicting patient survival outcomes using retinal imaging and neuro-ophthalmic data (accuracy: 78%).
- Conducted EDA, managed feature selection (RNFL, GCC), and validated model with leave-one-out cross-validation.
- Analyzed feature importance, created visualizations, and authored the ML methodology section of the publication (PMID: 39987704, PMCID: PMC11911018).
- Tools: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Work Experience

Mathematics & English Teacher, KBA Global Pvt. Ltd, Nepal (Nov 2021 – Feb 2023)

- Created a supportive, rigorous SAT and IELTS preparatory program
- Developed instructional strategies, materials and methodologies to create practical and enjoyable lessons for all levels of students
- Designed and produced new teaching materials as well as adapted existing ones
- Organized weekly mock test and evaluated students' progress to provide feedback on strengths and weaknesses
- Encouraged positive learning environment and an excellent student teacher interaction
- Aided other classroom teachers by designing special teaching programs for students with English Language difficulties
- Skillfully balanced the expectations of parents, students, and management to accommodate desires of all parties

Data Analyst Intern, Trobus Technologies, LLC (Laurel, Maryland, USA) (Sept 2020 – July 2021)

- Maintained KPI reports to analyze weekly, monthly cycle counts and identify trends.
- Assembled and analyzed data for business reports and created summary reports.
- Analyzed large data sets using SQL and excel. Elicit requirements from business users and designed interactive BI Dashboards using Power BI
- Created Power BI data extracts for sending reporting to management.
- Digitized investor's database by using Excel Power Pivot and Power Query, and monitored the consistency, accuracy, and continuity of data.
- Developed and maintained complex Excel spreadsheets with macro formulas, and custom formatted reports.
- Consult with business and technology partners on data visualization options and best practices